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Adventure Works: BI Dashboard Case Study

Introduction & Objectives

In modern enterprise environments, transforming raw transactional data into actionable BI is critical for strategic decision-making. This experiment focuses on designing and implementing analytical Key Performance Indicators (KPIs), custom DAX calculations, performance tuning via DAX Studio, and an interactive reporting dashboard using the Adventure Works dataset.

Our objectives are to:

- Develop custom DAX measures spanning core financials, aggregations, ranking, and time-based intelligence.
- Validate measure execution and query performance using DAX Studio.
- Create a Power BI dashboard.

Implementation

Uploaded AdventureWorks Dataset. Created a separate table “Measures Table” for storing “measures” made using DAX.

The measures I created are as follows:

-
```
Total Return Quantity =
SUM(
    'AdventureWorks Returns Data'[ReturnQuantity]
)
```
-
```
Total Returns =
COUNT(
    'AdventureWorks Returns Data'[ReturnQuantity]
)
```
-
```
Total Quantity =
SUM(
    'AdventureWorks Sales Data'[OrderQuantity]
)
```
-
```
Total Profit = [Total Sales] - [Total Cost]
```
-
```
Total Orders =
DISTINCTCOUNT(
    'AdventureWorks Sales Data'[OrderNumber]
)
```
-
```
Total Customers = DISTINCTCOUNT(
    'AdventureWorks Customer Lookup'[CustomerKey]
)
```

- ```
  Sales Variance = [Total Sales] - [Previous Month Sales]
  ```

- ```
  Total Cost =
  SUMX(
      'AdventureWorks Sales Data',
      'AdventureWorks Sales Data'[OrderQuantity]
      * RELATED('AdventureWorks Product Lookup'[ProductCost])
  )
  ```

- ```
  Sales Growth % =
  DIVIDE(
      [Sales Variance],
      [Previous Month Sales]
  )
  * 100
  ```

- ```
  Profit Margin % =
  DIVIDE(
      [Total Profit],
      [Total Sales]
  ) * 100
  ```

- ```
  Product Sales Rank =
  RANKX(
      ALL(
          'AdventureWorks Product Lookup'[ProductName]
      ),
      [Total Sales],
      ,
      DESC,
      Dense
  )
  ```

- ```
  Previous Month Sales =
  CALCULATE(
      [Total Sales],
      DATEADD(
          'AdventureWorks Calendar Lookup'[Date],
          -1,
          MONTH
      )
  )
  ```

- ```
  High Value Orders =
  CALCULATE(
      [Total Orders],
      FILTER(
          'AdventureWorks Sales Data',
          'AdventureWorks Sales Data'[OrderQuantity] > 5
      )
  )
  ```

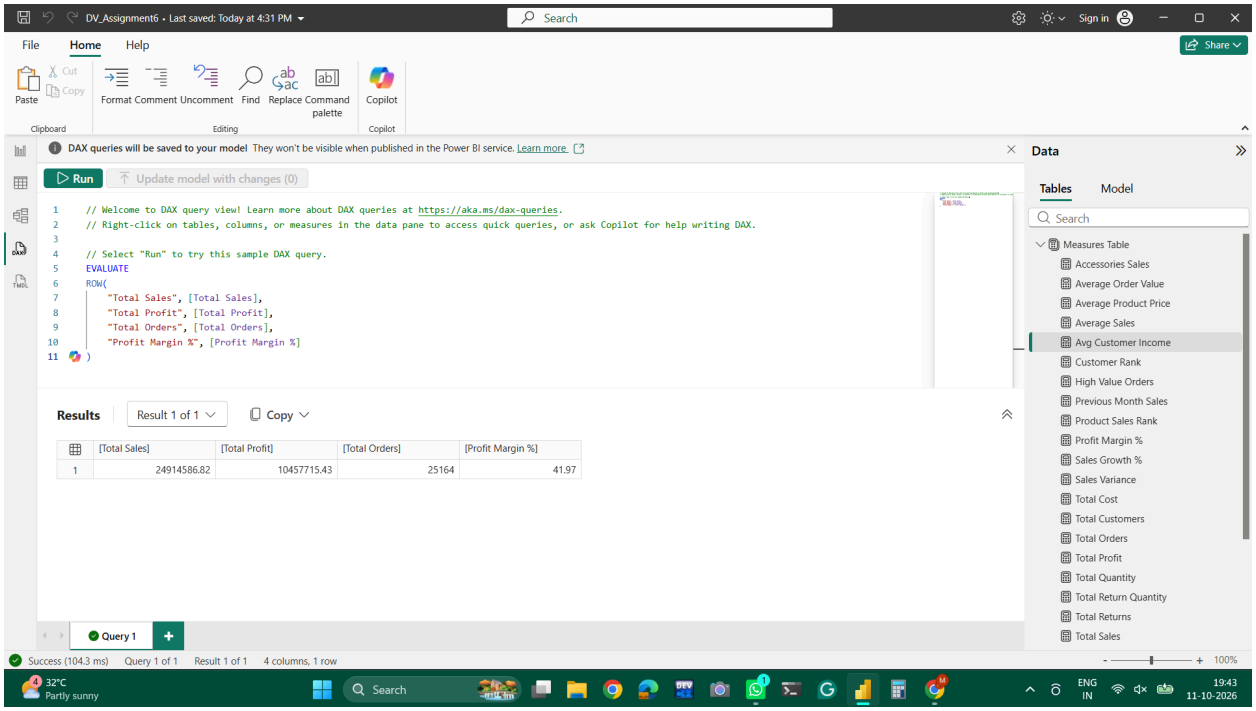
- ```
  Customer Rank =
  RANKX(
      ALL(
          'AdventureWorks Customer Lookup'[CustomerKey]
      ),
      [Total Sales],
      ,
      DESC,
      Dense
  )
  ```

- ```
  Avg Customer Income = AVERAGE('AdventureWorks Customer Lookup'[AnnualIncome])
  ```

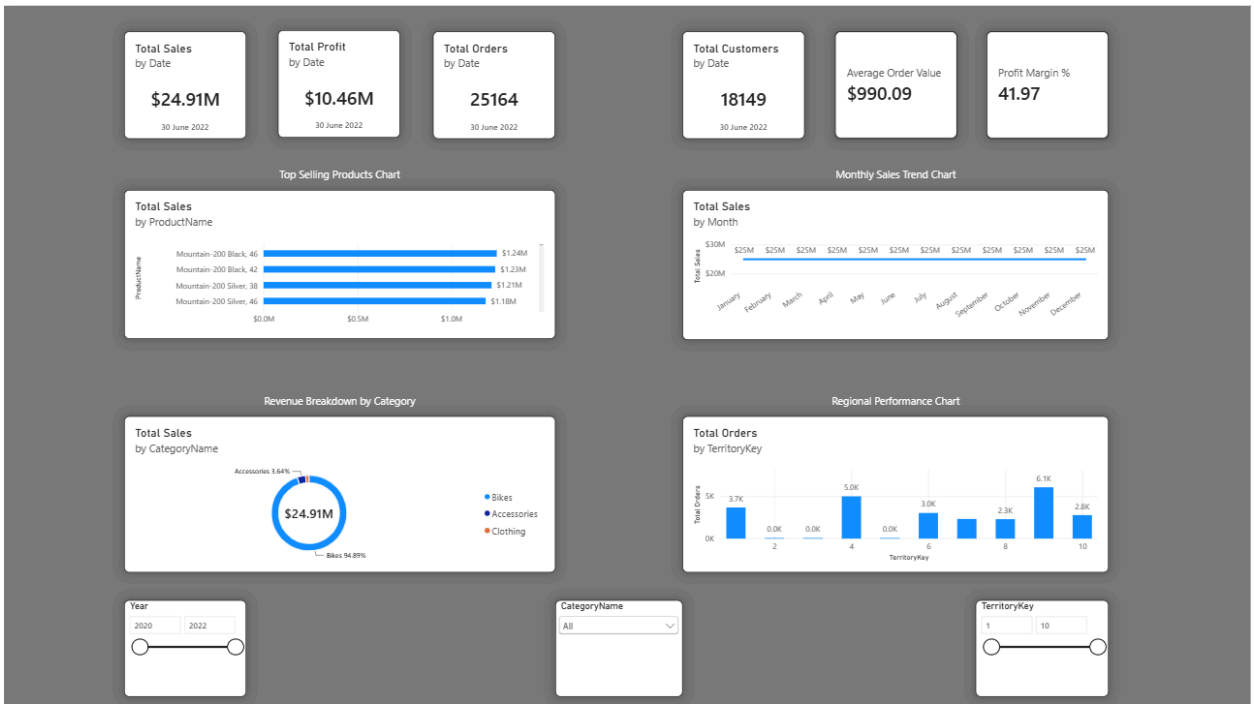
- ```
  Average Sales =
  AVERAGE(
      'AdventureWorks Product Lookup'[ProductPrice]
  )
  ```

- ```
  Average Order Value =
  DIVIDE(
      [Total Sales],
      [Total Orders]
  )
  ```
- ```
  Accessories Sales =
  CALCULATE(
      [Total Sales],
      'AdventureWorks Product Categories Lookup'[CategoryName] = "Accessories"
  )
  ```
- ```
  Average Product Price =
  AVERAGE(
      'AdventureWorks Product Lookup'[ProductPrice]
  )
  ```

Then I used DAX studio and was successfully able to use the query in the screenshot below:


  ```

And then I created a Dashboard as shown in below screenshot:



- I placed my KPIs at the very top.
    - Total Sales with Date
    - Total Profit with Date
    - Total Orders with Date
    - Total Customers with Date
    - Average Order Value
    - Profit Margin %
- Then the Stacked Bar chart which shows Top-selling products.(Y axis: ProductName, X axis: Total Sales)
- Beside that I placed a line chart showing monthly sales trends (X axis: TerritoryKey, Y axis: Total Orders).
- On the 2nd row I placed a Doughnut chart which shows revenue distribution across product categories (Legend: CategoryName, Value: Total Sales).
- Beside that there's a Clustered Column Chart that shows territory performance (X axis:TerritoryKey, Y axis: Total Orders).
- I have placed Slicers of Year, CategoryName, and TerritoryKey at the last row.

Analytical Findings & Conclusion

From above observations I have come to the conclusion that good pricing and clear seasonal sales trends help the business stay profitable.

LINKS TO BLOG

Platform	Link
LinkedIn	https://lnkd.in/p/dsFB6-qg
Google Docs	https://docs.google.com/document/d/1GtED9ff9dxW6V8YE_NqsXhlLe-Eq-m5a8NS_VidXvqo/edit?usp=sharing
GitHub	https://github.com/mst-2005/DataVisualization